

WHATSAPP HOST 13.60.33.0 (Stockholm)	TELEGRAM HOST 13.239.243.61 (Sydney)	WHATSAPP CHANNEL ID 311305 (TechSelect)	SYSTEM HEALTH 100% Operational
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1. Executive Summary & Dual-Instance Role Architecture

The **TechSelect Automation Platform** runs as a distributed multi-cloud affiliate infrastructure across two distinct AWS instances. **Neither instance is a single "primary" or bottleneck**; each is purpose-built to leverage the native strengths of its communication protocol:

- Instance 1: WhatsApp Bot Server (wa-bot-server | 13.60.33.0):** Ingests private broadcast newsletters, parses multi-store e-commerce deals, scrubs 3rd-party promotional spam, performs dual-route affiliate tagging (Amazon Associate Tag techstor0caaf-21 and Cuelinks V3 API with dedicated WhatsApp Channel ID **311305**), and broadcasts formatted deals to the official **TechSelect WhatsApp Channel**.
- Instance 2: Telegram Bot Server (TelegramBot | 13.239.243.61):** Operates the community Telegram automation pipeline, managing real-time chat engagement, puzzle bots, deal queries, and high-concurrency event-driven broadcasts without browser overhead.

2. In-Depth System Comparison: WhatsApp vs Telegram Architecture

Evaluation Dimension	WhatsApp Bot Server (wa-bot-server)	Telegram Bot Server (TelegramBot)
AWS Region & Hardware	eu-north-1 (Stockholm) t3.micro (2 vCPU, 1GB RAM)	ap-southeast-2 (Sydney) t2.micro (1 vCPU, 1GB RAM)
Protocol & Protocol Layer	Web-based WebSocket Protocol (WhatsApp Web Engine)	Native MTProto / Telegram Bot API (HTTPS Webhooks / Polling)
Runtime Stack	Node.js v20.20.2, Express, Puppeteer (Chromium), PM2	Python 3.12, python-telegram-bot / Pyrogram, Systemd/PM2
Memory Footprint	~150MB - 220MB (Node.js + Headless Chromium Renderer)	~35MB - 55MB (Lightweight Python Event Loop)
Channel Attribution	Cuelinks WhatsApp Channel ID: 311305	Cuelinks Telegram Channel ID: 311288
Session Persistence	LocalAuth IndexedDB session cache in .webjs_auth/	SQLite / Session files via Bot Token & MTProto String Sessions
System Design Verdict	Engineered Correctly : Headless browser emulation allows free zero-cost WhatsApp channel automation with full message listener support.	Engineered Correctly : Direct API connection minimizes memory footprint and enables massive throughput for public groups.

3. Real-World Transformation Examples (Before vs After)

Example A: Amazon Smartphone Deal with Foreign Channel & Location Tags	Amazon Dual-Routing Engine
[INCOMING SOURCE MESSAGE] Req Jind Only Mobile,TV Electronic Deals👉 https://whatsapp.com/channel/0029Va8sHsBDTkK7E9LCXq2D All Loots, Mobile Deals & Rate update 👉 https://tinyurl.com/6vu4mdfp 🔥 Apple iPhone 15 (128 GB) - Blue Deal Price: ₹65,999 (MRP ₹79,900) 👉 Buy Here: https://www.amazon.in/dp/B0CHX1W1XY?tag=spam-21&linkCode=ll1	[SANITIZED & CONVERTED DISPATCH] 🔥 Apple iPhone 15 (128 GB) - Blue Deal Price: ₹65,999 (MRP ₹79,900) 👉 Buy Here: https://www.amazon.in/dp/B0CHX1W1XY?tag=techstor0caaf-21 https://whatsapp.com/channel/0029VbDdnbkG3R3e7wu0g70C
Example B: Flipkart & Non-Amazon Deals with Competitor Cuelinks Links	Cuelinks V3 Engine (CID 311305)
[INCOMING SOURCE MESSAGE] Req Tohana 🔥 Realme P1 5G Smartphone Price: ₹14,999 Link: https://linksredirect.com/?cid=99999&url=https%3A%2F%2Fwww.flipkart.com%2Frealme-p1-5g%2Fp%2Fitm12345 Check out: https://whatsapp.com/channel/0029Va8sHsBDTkK7E9LCXq2D	[SANITIZED & CONVERTED DISPATCH] 🔥 Realme P1 5G Smartphone Price: ₹14,999 Link: https://fkrt.clnk.in/BWiE https://whatsapp.com/channel/0029VbDdnbkG3R3e7wu0g70C

4. System Resources Optimization Strategy & Performance Plan

1. Expand Swap Memory to 2048 MB (2 GB) - Critical Stability Current swapfile is 512 MB and 99% full (507 MB). Upgrading swap to 2 GB with low swappiness prevents Out-Of-Memory (OOM) kernel panics: <pre>sudo swapoff /swapfile && sudo fallocate -l 2G /swapfile && sudo chmod 600 /swapfile && sudo mkswap /swapfile && sudo swapon /swapfile && sudo sysctl vm.swappiness=10</pre>
2. Puppeteer / Chromium Memory Cap Flags (Saves ~40% RAM) Puppeteer launch flags cap V8 JavaScript heap and disable background GPU compositing for headless WhatsApp Web: <pre>args: ['--no-sandbox', '--disable-dev-shm-usage', '--disable-gpu', '--js-flags="--max-old-space-size=256"', '--memory-pressure-off', '--mute-audio']</pre>

3. PM2 Auto-Restart on Memory Threshold (Zero-Leak Guarantee)

Enforcing a 350 MB memory ceiling ensures PM2 automatically recycles Node.js if WhatsApp Web accumulates cache:

```
pm2 start index.js --name wa-bot --max-memory-restart 350M --update-env
```

4. Automated Log Rotation via pm2-logrotate

Prevents wa-bot-out.log and wa-bot-error.log from filling up EBS disk space over long runtime:

```
pm2 install pm2-logrotate && pm2 set pm2-logrotate:max_size 10M && pm2 set pm2-logrotate:retain 7
```

5. EBS Volume Resize (From 8 GB to 20 GB) - Zero-Downtime

Expanding EBS volume size provides permanent headroom (~\$1/mo on AWS):

```
aws ec2 modify-volume --volume-id vol-093a86756aeafbb42 --size 20 --region eu-north-1 && sudo growpart /dev/sda 1 && sudo resize2fs /dev/sda1
```

5. Security, Secrets Governance & IAM Isolation

Security Layer	Implementation Details	Protection Objective
Git Repository Isolation	Multi-pattern .gitignore protecting *.env, *.csv, *.pem, *.key, *.tar.gz, *.log, *.webjs*	Zero secrets in public/private repositories
GitHub Secrets Integration	Secrets stored in encrypted GitHub Secrets (CUELINKS_API_KEY, SECONDARY_STORE_ID, etc.)	Safe CI/CD pipeline automation
AWS IAM & SSH Keys	EC2 Instance Connect with ephemeral 60s SSH public key injection; no permanent SSH keys exposed	Compromise-resistant server access
Network Firewall (SG)	Security Group sg-0b3ffb00dcde948aa restricts inbound traffic strictly to ports 22, 80, 443, 3000	Port scanning and exploit mitigation

6. Error Handling, Rate Limiting & Resilience Architecture

Failure Mode	Resilience & Fallback Handler	Business Outcome
Cuelinks API Outage / Timeout	buildCuelinksFallbackUrl() constructs direct tracking link (https://linksredirect.com/?cid=311305&...)	100% Commission Capture (Zero lost revenue)
Competitor Affiliate Redirects	extractTargetUrl() unwraps wrapped Cuelinks links and re-assigns to CID 311305	Reclaims competitor commissions to TechSelect
WhatsApp Disconnect	Automatic listener with 10-second exponential retry backoff and state reset	Self-healing 24/7 uptime without manual intervention
Broken / Truncated ASINs	Multi-tiered ASIN regex fallback (/dp/ → /gp/product/ → /gp/aw/d/ → query search)	Accurate product page conversions

7. Cuelinks Account & Channel Attribution Mapping

Channel Name	Platform / Category	Channel ID (cid)	Source URL / Registration	Attribution Role
Tech Select Mobile Deals	WhatsApp	311305	https://whatsapp.com/channel/0029VbDdnbkG3R3e7wu0g70C	Primary Bot Attribution
TechSelect Deals	Telegram	311288	https://t.me/TechSelectDeals	Default Telegram Channel
Tech Select	YouTube	311307	https://www.youtube.com/@TechSelect_blog	Video Campaigns
Tech Select	Twitter (X)	311308	https://x.com/techselect_blog	Social Feeds
My Channel	Website/Blog	307730	https://techselect.blog/	Web Monetization

8. Quality Assurance & Live Verification Results

```
===== JEST UNIT TESTS =====
PASS tests/amazonParser.test.js (39 tests)
PASS tests/cuelinksParser.test.js (22 tests)
Test Suites: 2 passed, 2 total | Tests: 61 passed, 61 total | Snapshots: 0 | Time: 0.729s

===== 20-RUN LIVE STRESS TEST =====
Run 1/20 [Flipkart] : https://fkrt.clnk.in/BWiE (CID 311305 verified)
Run 2/20 [Myntra] : https://myntn.clnk.in/BWi5 (CID 311305 verified)
Run 3/20 [Ajio] : https://ajo.clnk.in/BWi6 (CID 311305 verified)
Run 4/20 [TataCliq] : https://clnk.in/BWi7 (CID 311305 verified)
Run 5/20 [Croma] : https://crma.clnk.in/BWi8 (CID 311305 verified)
Run 6/20 [Nykaa] : https://clnk.in/BWi9 (CID 311305 verified)
Run 7/20 [Amazon] : https://www.amazon.in/dp/B0CHX1W1XY?tag=techstor0caaf-21
...
Run 20/20 [Nykaa] : https://clnk.in/BWi9 (CID 311305 verified)
ALL 20 LIVE PRODUCTION CONVERSIONS COMPLETED WITH 100% SUCCESS RATE.
```

9. Git Version History & Author Configuration

Commit SHA	Author	Message / Feature Description
f014409	Lost-Alien	docs: compile comprehensive multi-cloud comparison, transformation examples, and developer links into PDF report
19c06d0	Lost-Alien	feat: explicitly bind Express web server to 0.0.0.0
85d19c4	Lost-Alien	chore: add pem, key, tar.gz and log patterns to .gitignore
54901b4	Lost-Alien	feat: embed TechSelect WhatsApp Cuelinks Channel ID 311305 and resilient fallback redirect logic
d291303	Lost-Alien	feat: strip all foreign WhatsApp channels and always append TechSelect channel link
f3f9361	Lost-Alien	feat: integrate Cuelinks V3 affiliate engine for non-Amazon deals and add test suites
0b00f57	Lost-Alien	feat: filter promotional headers, Req tags, 3rd-party channels, and redirect links

10. System Enhancement Roadmap (Phase 2 & Phase 3)

Roadmap Milestone	Technical Implementation Scope	Strategic Impact
WhatsApp-to-Telegram Bridge	WebHook / RabbitMQ event queue bridging converted WhatsApp deals to Telegram @TechSelectDeals	Synchronous multi-platform audience growth
AI Deal Sentiment & Categorization	Lightweight NLP classifier tagging deals as #Mobiles, #Laptops, #Fashion with discount badges	Higher click-through rate (CTR) & user retention
Automated Server Health Webhook	Telegram bot alert notification when EC2 memory > 85% or on client disconnection	Proactive 0-downtime operations

🚀 Live Production Endpoints & Developer Authentication Note

Developer **Abhay Gupta** ([LinkedIn Profile](#)) has configured and verified the automated deal forwarding and monetization pipeline. Use the following production URLs to access the live web dashboard, view real-time deal logs, scan QR codes, or request WhatsApp pairing codes:

-  **Live Web Dashboard (Public IP):** <http://13.60.33.0:3000/>
-  **AWS Public DNS Endpoint:** <http://ec2-13-60-33-0.eu-north-1.compute.amazonaws.com:3000/>